GREENAUDIT – HOME ENERGY AUDIT WEBSITE

Mini Project Documentation

Project Title: GreenAudit – Home Energy Audit Website

Department: Electrical and Electronics Engineering (EEE)

Academic Level: B.Tech 2nd Year

Development Platform: Base44 AI

Developed By: Anusha

Project Link:

https://preview-sandbox--6ac9d825b13f2caa1737bb01.base44.app/

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1. ABSTRACT

GreenAudit is a web-based Home Energy Audit System designed to help households understand their electricity consumption and reduce unnecessary energy usage. The website allows users to enter household appliance details, including power rating, quantity, and daily operating hours.

Based on the entered information, the system estimates daily and monthly electricity consumption, calculates approximate electricity charges, and identifies appliances that consume more energy. It also provides useful energy-saving recommendations.

The project is developed using the Base44 AI platform. It demonstrates the practical application of electrical energy calculations and promotes energy conservation and environmental awareness.

2. INTRODUCTION

Electricity is an essential part of everyday life. Household appliances such as fans, lights, televisions, refrigerators, and air conditioners consume electrical energy.

Many people do not know how much electricity individual appliances consume. As a result, electricity may be wasted, and monthly bills may increase.

GreenAudit provides a simple website to estimate household electricity consumption. It helps users understand energy usage, estimate electricity charges, and identify possible ways to save energy.

3. PROBLEM STATEMENT

Households often lack an easy method to calculate appliance-wise electricity consumption and estimate the corresponding electricity cost.

Manual calculations can be confusing, and users may not know which appliances consume the most energy.

Therefore, GreenAudit is proposed as a web-based solution to estimate energy consumption, analyze appliance usage, and provide energy-saving suggestions.

4. OBJECTIVES

The main objectives of GreenAudit are:

1. To calculate estimated daily electricity consumption.
2. To estimate monthly energy consumption in units.
3. To calculate approximate monthly electricity charges.
4. To identify high-energy-consuming appliances.
5. To provide practical energy-saving recommendations.

6. To promote energy efficiency and environmental awareness.

5. PROPOSED SYSTEM

GreenAudit is a responsive website that accepts household appliance details and processes them using energy calculation formulas.

The proposed system includes:

- Appliance details entry.
- Daily and monthly energy calculation.
- Electricity bill estimation.
- Appliance-wise energy comparison.
- Energy-saving recommendations.
- Sample household data for demonstration.

The basic project works using user-entered information and does not require physical sensors or IoT hardware.

6. SYSTEM ARCHITECTURE

The system follows this workflow:

User Input → Energy Calculation → Bill Estimation → Dashboard → Energy-Saving Recommendations

Working Process

1. The user enters appliance details.
2. The website calculates energy consumption.

3. Daily and monthly consumption are estimated.

4. The electricity charge is calculated using the entered tariff.

5. Appliance-wise energy usage is compared.

6. The website displays results and energy-saving suggestions.

7. MAIN FEATURES

7.1 Home Page

The home page introduces the GreenAudit project and explains its purpose. It provides navigation to the energy audit and other website features.

7.2 Energy Audit

Users can enter the following details for household appliances:

- Appliance name.
- Rated power in watts.
- Number of appliances.
- Daily operating hours.

These details are used to estimate electricity consumption.

7.3 Energy Dashboard

The dashboard can display:

- Estimated daily energy consumption.
- Estimated monthly energy consumption.
- Approximate monthly electricity charges.
- Appliance with the highest energy consumption.
- Appliance-wise energy usage charts.

7.4 Electricity Bill Calculator

The user enters the electricity tariff per unit. The website estimates the monthly energy charge based on calculated monthly consumption.

7.5 Energy-Saving Recommendations

The website can suggest practical actions, such as:

- Replacing traditional bulbs with LED bulbs.
- Switching off unused lights and fans.
- Reducing unnecessary air-conditioner usage.
- Choosing energy-efficient appliances.
- Reducing unnecessary operating hours.

7.6 Green Score

The website may display an indicative energy-efficiency score to encourage users to improve their energy-saving habits. This score is a project metric, not an official energy rating.

8. WORKING PRINCIPLE AND FORMULAS

GreenAudit uses basic electrical energy calculations.

8.1 Daily Energy Consumption

Daily Energy (kWh) = Power (W) × Quantity × Usage Hours per Day ÷ 1000

8.2 Monthly Energy Consumption

Monthly Energy (kWh) = Daily Energy Consumption × 30

8.3 Estimated Monthly Electricity Charge

Estimated Bill (₹) = Monthly Energy Consumption × Tariff per Unit

8.4 Estimated Energy Savings

Energy Savings (kWh) = Power (W) × Quantity × Hours Reduced per Day × Number of Days ÷ 1000

Example Calculation

Consider one fan with a rated power of 60 W, operating for 8 hours per day.

Daily energy consumption:

60 × 1 × 8 ÷ 1000 = 0.48 kWh

Monthly energy consumption:

0.48 × 30 = 14.4 units

Assume the electricity tariff is ₹6 per unit.

Estimated monthly energy charge:

14.4 × 6 = ₹86.40

This is an illustrative energy-charge calculation. Actual electricity bills may include fixed charges, taxes, and slab-based tariffs.

9. TECHNOLOGIES USED

Base44 AI Platform

Base44 is used to generate and build the website from natural-language instructions.

Web Technologies

The project applies website development concepts such as:

- HTML for page structure.
- CSS for layout and styling.
- JavaScript concepts for calculations and interactive behaviour.
- Charts and dashboard components for data visualization.

The exact implementation technologies depend on the code and configuration generated by Base44.

10. METHODOLOGY

The project development process involves the following steps:

1. Identify the problem of household electricity wastage.
2. Define the objectives and website requirements.
3. Prepare a prompt describing the required features.
4. Generate the website using Base44 AI.
5. Configure appliance inputs and energy calculations.
6. Design the dashboard and recommendations.
7. Test the website with sample values.
8. Publish the website and share the project link.

11. TESTING

The following checks can be used to verify the website.

Test| Input| Expected Result

Daily energy| 60 W, 8 hours| 0.48 kWh/day

Monthly energy| 0.48 × 30| 14.4 units

Energy charge| 14.4 units at ₹6/unit| ₹86.40

Appliance entry| Add an appliance| Details are added

Input validation| Negative power value| Invalid input should be rejected

Dashboard| Change appliance details| Results should update accordingly

These are expected results for testing. Mark each feature as working only after checking it on the actual website.

12. ADVANTAGES

- Easy to use and understand.
- Helps estimate household energy consumption.
- Identifies appliances with high energy usage.
- Helps users understand approximate electricity costs.
- Encourages electricity conservation.
- Demonstrates practical applications of EEE concepts.
- Can be demonstrated without physical hardware.

13. LIMITATIONS

- Results depend on the accuracy of user-entered values.
- Monthly consumption is estimated using a 30-day period.
- Estimated bills may differ from actual electricity bills.
- The basic system does not directly measure real-time energy consumption.
- Energy-saving suggestions provide estimates rather than guaranteed savings.

14. FUTURE SCOPE

The project can be enhanced with the following features:

1. Integration with smart meters and IoT sensors.
2. Real-time electricity consumption monitoring.
3. Historical energy usage analysis.
4. Downloadable home energy audit reports.
5. Electricity bill upload and analysis.
6. Detailed electricity tariff calculations.

7. Personalized recommendations based on usage history.

15. CONCLUSION

GreenAudit – Home Energy Audit Website provides a simple digital method for estimating household electricity consumption and energy charges. It helps users understand appliance-wise energy usage and explore practical methods to reduce unnecessary electricity consumption.

The project applies basic electrical engineering concepts to a real-world problem and promotes energy efficiency and environmental awareness.

GreenAudit demonstrates how an AI-assisted website-building platform such as Base44 can be used to develop an educational engineering mini project.

16. PROJECT LINK

GreenAudit – Home Energy Audit Website

https://preview-sandbox--6ac9d825b13f2caa1737bb01.base44.app/

17. REFERENCES

1. Base44 AI App Builder – https://base44.com/

2. GreenAudit project website – https://preview-sandbox--6ac9d825b13f2caa1737bb01.base44.app/

3. Basic electrical engineering concepts of electrical power and energy consumption.

Project Slogan:

“Save Energy Today, Secure Tomorrow.”

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